# Machine learning for sports betting

NBA and English Premier League: accuracy, calibration and archived-price returns.

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Statistical Arbitrage | Final project | 23 October 2026

PART 1

## Methodology

The shared prediction-to-betting framework

PART 2

## NBA results

Aman Dhillon

PART 3

## Premier League

Panagiotis Housos

Sources: ML_Sports_Betting_Project_Proposal.pdf | Full paths in notes

# Common methodology: from past games to bets

The proposal compares calibration-based and accuracy-based model selection. Implementation differs where the sports require it.

### 1 Earlier games

Earlier team stats
Prior-season strength
Player/xG extensions

### 2 Probabilities

NBA: home / away
EPL: home / draw / away
Exclude target inputs

### 3 Model selection

Accuracy branch
Calibration-error branch
Later-season evaluation

### 4 Betting rule

Compare probabilities with archived odds
Apply explicit stake rules

**Estimated return per unit staked = model probability × decimal odds − 1**

We test whether better probability estimates improve betting outcomes.
A higher classification accuracy or smaller calibration error does not guarantee a profitable price.

Sources: Data.ipynb; METHODOLOGY.md; ML_Sports_Betting_Project_Proposal.pdf | Full paths in notes

# Common metrics; explicit league differences

Compare outcomes within each league. Keep the financial denominators and funding conventions separate.

| Setting | NBA implementation | Premier League implementation |
|---|---|---|
| Outcome space | Home / away | Home / draw / away |
| Calibration criterion | 20 bins; score set to 1 if fewer than 16 bins are occupied | Mean classwise ECE over 3 classes; 10 bins; no occupancy penalty |
| Post-fit calibration | No separate held-out temperature fit | Both initial branches receive held-out temperature scaling |
| Stake and starting cash | 100 fixed or 1/8 Kelly; start 10,000 | 1,000 fixed or capped Kelly;<br>start 100,000 in expanded accounts |
| Funding | Game-by-game settlement | Joint date-start cash allocation before settlement |

**Capital return = profit / initial cash.** **Stake ROI (yield) = profit / total money staked.**

The NBA source calls capital return “ROI”. We relabel it here. Forecast accuracy is also different from the hit ratio of selected bets.

Sources: Data.ipynb; metrics.py; METHODOLOGY.md | Full paths in notes

# NBA: data and model selection

Team box-score differences are lagged; each team's first ten appearances are excluded as prediction targets.

**ELIGIBLE GAME ROWS**

**14,471**

Across the supplied historical sample

**FEATURE SCREENING**

**26 → 25 → 9**

Candidate → drift-screened → nonredundant

**BETTING EVALUATION**

**1,074**

2018/19 games with odds

| Selection branch | Chosen model | Selected inputs | 2017/18 selection criterion |
|---|---|---|---|
| Calibration | Logistic regression | 2 | Penalised ECE 4.31% |
| Accuracy | SVM | 4 | Accuracy 66.33% |

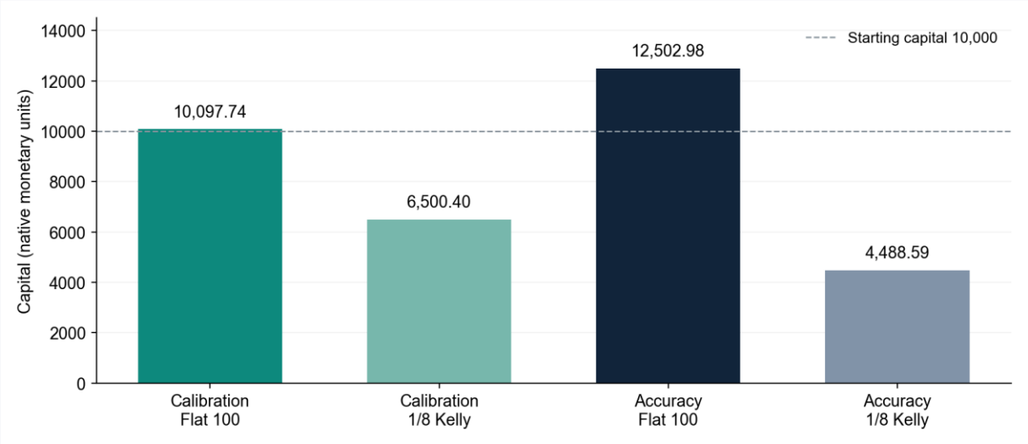
Train: 2014/15–2015/16 → tune: 2016/17 → family selection: 2017/18 → bet: 2018/19.
The two branches use different selected feature sets; the row scores are not joint metrics of one fitted model.

Sources: Data.ipynb | Full paths in notes

# NBA: saved 2018/19 betting results

Four rules, each starting with 10,000. Fixed 100 stakes equal 1% of initial capital.

| Branch / model | Stake rule | Final capital | Capital return | Trade frequency | Hit ratio |
|---|---|---|---|---|---|
| Calibration (LR) | 100 fixed | 10,097.74 | 0.98% | 85.3% | 38.0% |
| Calibration (LR) | 1/8 Kelly | 6,500.40 | -35.00% | 85.3% | 38.0% |
| Accuracy (SVM) | 100 fixed | 12,502.98 | 25.03% | 88.6% | 34.9% |
| Accuracy (SVM) | 1/8 Kelly | 4,488.59 | -55.11% | 88.6% | 34.9% |




Home value is checked first.
Otherwise check away value.
Kelly uses 1/8 of full Kelly,
limited to available cash.
Both saved Kelly outcomes lose.

Saved notebook results: LR reproduced to cents. SVM replay did not match; its saved results are reported, not independently verified.

Sources: Data.ipynb; nba_evidence.json | Full paths in notes

# NBA: nine-season annual robustness

Each season starts a fresh 10,000 account with fixed 100 requests. Annual paths are not one funded continuous portfolio.

**CALIBRATION-BRANCH MEAN**

**-17.34%**

Mean annual capital return

**ACCURACY-BRANCH MEAN**

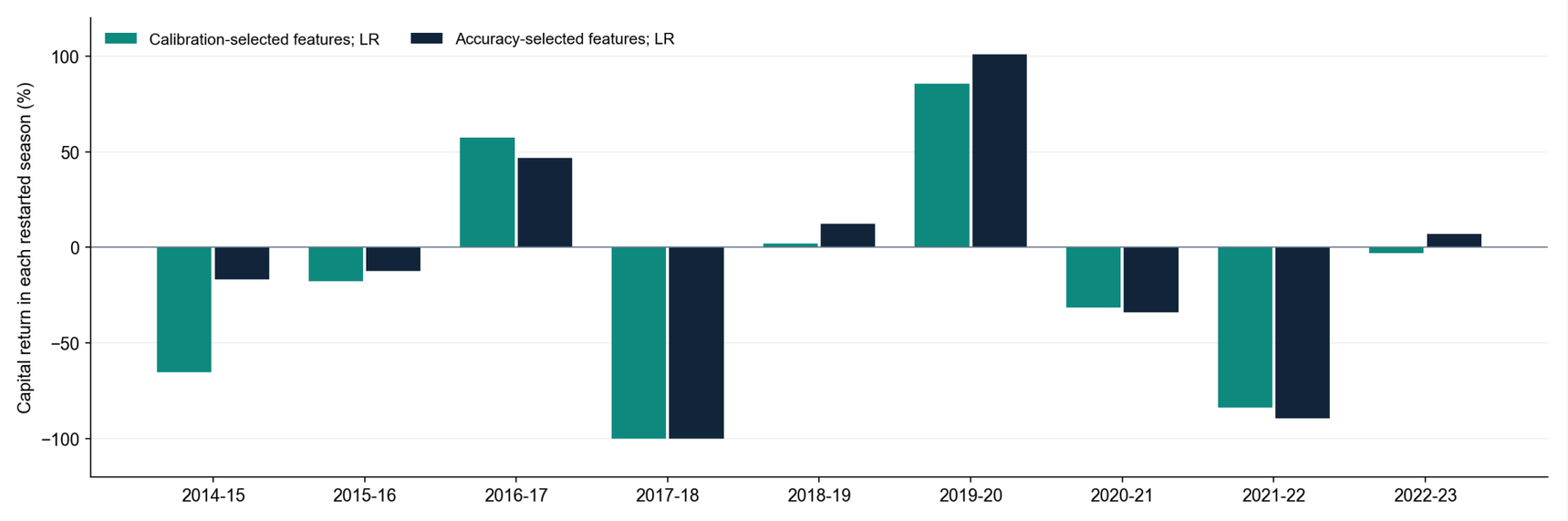
**-9.49%**

Mean annual capital return

**CALIBRATION WINS**

**3 / 9**

Both branches use logistic regression here




This extension reselects features but does not repeat the four-family search. The annual variation does not support a dependable profit advantage.

# EPL: the proposal's accuracy–calibration comparison

The first experiment evaluates 374 matches in 2025/26 after separate tuning, calibration, family selection and strategy validation.

| Probability model | Accuracy | ECE 10 | Log loss |
|---|---|---|---|
| Accuracy-selected SVM | 48.13% | 6.20% | 1.0763 |
| Calibration-selected MLP | 44.39% | 4.16% | 1.0564 |
| Bet365 market benchmark | 48.66% | 4.36% | 1.0191 |

PRIMARY POLICIES

**Hold cash**

Neither branch found an admissible positive-return validation threshold.

## SECONDARY DIAGNOSTIC: predeclared 5% edge, best archived quotes, 1,000 flat stakes from 100,000

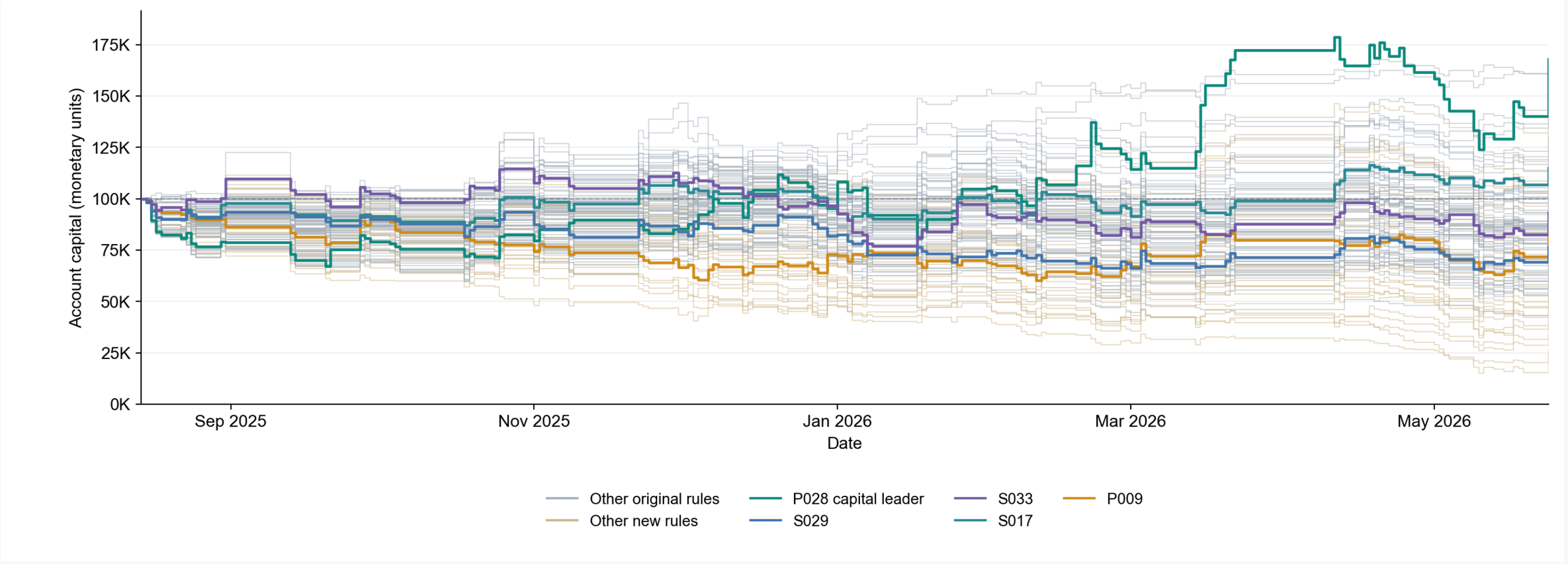
| Model | Bets | Profit | Stake ROI | Capital return |
|---|---|---|---|---|
| SVM | 371 | -8,330 | -2.25% | -8.33% |
| MLP | 333 | 1,370 | 0.41% | 1.37% |

The MLP's +0.41% stake ROI is uncertain: its conditional 95% interval includes zero. The following EPL extensions are exploratory.

Sources: RESULTS.md; classification_scores.csv; metrics.csv | Full paths in notes

# EPL: money progression of all 116 strategies

2025/26 is the common season available for every account. Each starts with 100,000 monetary units.




All 84 original and 32 new rules are included. The next model pages label their separate four-year statistics explicitly.

Sources: graph_equity_2025_all.csv | Full paths in notes

# EPL: best prediction model: odds + team/player form

Lowest prediction error among the learned models on the same four-year sample. Trading statistics: 2022/23–2025/26, 1,470 matches.

## WHAT IT DOES

Starts with bookmaker probabilities, then adjusts them using earlier team results and player form.

**ENDING CAPITAL**

**123,340**

Started at 100,000

**RETURN ON INITIAL CAPITAL**

**23.34%**

Four-year account

## WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 54.15% | Prediction loss: 0.9693

| Trading statistic | Result |
|---|---|
| Bets / trading frequency | 1,147 / 78.03% |
| Hit ratio: bets won | 31.04% |
| Expected gain per trade | 852.95 |
| Expected loss per trade | 663.70 |
| Profit / ROI on total stakes | 23,340 / 2.03% |
| Largest drop from a prior peak | 31.88% |

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: predictions.parquet; metrics_continuous.parquet | Full paths in notes

# EPL: simplest model: past result frequencies

No match-specific inputs: every game in a season gets the same three probabilities. Trading statistics: 2022/23–2025/26, 1,470 matches.

## WHAT IT DOES

Counts how often home wins, draws and away wins happened in the training data. Uses those fractions for every game.

## WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 44.29% | Prediction loss: 1.0700

**ENDING CAPITAL**

**29,760**

Started at 100,000

**RETURN ON INITIAL CAPITAL**

**-70.24%**

Four-year account

| Trading statistic | Result |
|---|---|
| Bets / trading frequency | 1,447 / 98.44% |
| Hit ratio: bets won | 23.01% |
| Expected gain per trade | 1739.52 |
| Expected loss per trade | 611.96 |
| Profit / ROI on total stakes | -70,240 / -4.85% |
| Largest drop from a prior peak | 89.49% |

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: predictions.parquet; metrics_continuous.parquet | Full paths in notes

# EPL: team model: earlier team performance

Uses team performance and data-coverage controls; no player-performance or odds predictors. Trading statistics: 2022/23–2025/26, 1,470 matches.

## WHAT IT DOES

Uses earlier goals, shots on target, corners, points, rest days and previous-season results to estimate home/draw/away chances.

## WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 52.93% | Prediction loss: 0.9876

**ENDING CAPITAL**

**119,790**

Started at 100,000

**RETURN ON INITIAL CAPITAL**

**19.79%**

Four-year account

| Trading statistic | Result |
|---|---|
| Bets / trading frequency | 1,241 / 84.42% |
| Hit ratio: bets won | 29.57% |
| Expected gain per trade | 962.54 |
| Expected loss per trade | 648.43 |
| Profit / ROI on total stakes | 19,790 / 1.59% |
| Largest drop from a prior peak | 40.35% |

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: predictions.parquet; metrics_continuous.parquet | Full paths in notes

# EPL: player model: earlier Fantasy Premier League form

Uses previous player records; no team-strength or bookmaker-odds predictors. Trading statistics: 2022/23–2025/26, 1,470 matches.

## WHAT IT DOES

Uses earlier FPL points, goals and assists.
Adds minutes and other player statistics.
Looks back over five and ten games.

## WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 50.07% | Prediction loss: 1.0218

**ENDING CAPITAL**

**0**

Started at 100,000

**RETURN ON INITIAL CAPITAL**

**-100.00%**

Four-year account

| Trading statistic | Result |
|---|---|
| Bets / trading frequency | 944 / 64.22% |
| Hit ratio: bets won | 25.00% |
| Expected gain per trade | 1098.04 |
| Expected loss per trade | 618.22 |
| Profit / ROI on total stakes | -100,000 / -10.87% |
| Largest drop from a prior peak | 100.00% |

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: model_metrics_pooled.csv; metrics_continuous.parquet | Full paths in notes

# Appendix: the four NBA betting rules

One bet at most per game; settle sequentially. This is not the EPL joint same-date funding rule.

| Forecast branch | Sizing rule | Money requested per trade | Entry rule |
|---|---|---|---|
| Calibration / LR | Fixed | 100 per bet = 1% of initial 10,000 | Home value first; otherwise away value |
| Calibration / LR | Eighth Kelly | 1/8 Kelly; variable; cap only available cash | Home value first; otherwise away value |
| Accuracy / SVM | Fixed | 100 per bet = 1% of initial 10,000 | Home value first; otherwise away value |
| Accuracy / SVM | Eighth Kelly | 1/8 Kelly; variable; cap only available cash | Home value first; otherwise away value |

**Eighth-Kelly stake = current cash × (1/8) × max(0, (probability × odds − 1) / (odds − 1)).**

Do not add an EPL-style 2% cap to these source rules. A positive estimated return is a model claim, not a guaranteed gain.

# EPL appendix: how much capital each rule uses

Stake means the amount requested for one bet. The next pages state the setting for every strategy.

| Stake type | Plain meaning | Example |
|---|---|---|
| Fixed 1,000 | Same requested amount each bet: 1% of INITIAL 100,000 | Cash 80,000 → still request 1,000 |
| 1% of current cash | One percent of cash at the start of the match date | Cash 80,000 → request 800 |
| Kelly sizing | Amount varies with estimated advantage and odds; the listed % is a ceiling | A 2% cap means at most 2,000 when cash is 100,000 |

Kelly example: 100,000 cash, a 40% chance and odds of 3.
Quarter Kelly requests 2,500; a 2% cap reduces the bet to 2,000.

Kelly stake = opening cash × min(cap, Kelly fraction × (probability × odds − 1) / (odds − 1)).

All bets on a date share opening cash. Date caps apply to their combined stakes. Reduce all requests proportionally when needed; never borrow.

Sources: engine.py; all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (1/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S001–S004 | Original SVM forecast | No bets; keep cash | 0 | No betting |
| S005 / S006 | Original SVM forecast | Estimated return > 5% | S005: 1,000<br>S006: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S007 / S008 | Original SVM forecast | Estimated return > 5% | S007: 1,000<br>S008: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| S009–S012 | Original neural-network forecast | No bets; keep cash | 0 | No betting |
| S013 / S014 | Original neural-network forecast | Estimated return > 5% | S013: 1,000<br>S014: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S015 / S016 | Original neural-network forecast | Estimated return > 5% | S015: 1,000<br>S016: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| S017 / S018 | Team form | Estimated return > 5% | S017: 1,000<br>S018: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S019 / S020 | Team form + chance quality | Estimated return > 5% | S019: 1,000<br>S020: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S021 / S022 | Team form + player form | Estimated return > 5% | S021: 1,000<br>S022: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (2/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S023 / S024 | Team + chance quality + player form | Estimated return > 5% | S023: 1,000<br>S024: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S052 / S055 | Team + chance quality + player form | Estimated return > 5% | S052: 1% of current cash<br>S055: 1/8 Kelly; ≤1% cash | Best odds<br>Cash available |
| S056 | Team + chance quality + player form | Estimated return > 5% | S056: 1/2 Kelly; ≤2% cash | Best odds<br>Cash available |
| S025 / S026 | Bookmaker odds + team form | Estimated return > 5% | S025: 1,000<br>S026: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S027 / S028 | Bookmaker odds + team/chance quality | Estimated return > 5% | S027: 1,000<br>S028: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S029 / S030 | Bookmaker odds + team/player form | Estimated return > 5% | S029: 1,000<br>S030: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S031 / S032 | Bookmaker odds + team/chance/player form | Estimated return > 5% | S031: 1,000<br>S032: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S053 / S057 | Bookmaker odds + team/chance/player form | Estimated return > 5% | S053: 1% of current cash<br>S057: 1/8 Kelly; ≤1% cash | Best odds<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (3/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S058 | Bookmaker odds + team/chance/player form | Estimated return > 5% | S058: 1/2 Kelly; ≤2% cash | Best odds<br>Cash available |
| S033 / S034 | Past home/draw/away frequencies | Estimated return > 5% | S033: 1,000<br>S034: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S035 / S036 | Bookmaker probabilities | Estimated return > 5% | S035: 1,000<br>S036: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S037 / S038 | Calibrated bookmaker probabilities | Estimated return > 5% | S037: 1,000<br>S038: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S039 | Bookmaker probabilities | Always home win<br>No return filter | S039: 1,000 | Best odds<br>Cash available |
| S040 | Bookmaker probabilities | Always draw<br>No return filter | S040: 1,000 | Best odds<br>Cash available |
| S041 | Bookmaker probabilities | Always away win<br>No return filter | S041: 1,000 | Best odds<br>Cash available |
| S042 | Bookmaker probabilities | Bookmaker's most likely result<br>No return filter | S042: 1,000 | Best odds<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (4/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S043 | Bookmaker probabilities | Bookmaker's least likely result<br>No return filter | S043: 1,000 | Best odds<br>Cash available |
| S044 | Team + chance quality + player form | Model's most likely result<br>No return filter | S044: 1,000 | Best odds<br>Cash available |
| S045 | Bookmaker odds + team/chance/player form | Model's most likely result<br>No return filter | S045: 1,000 | Best odds<br>Cash available |
| S046 / S047 | Average of four team/player forecasts | Estimated return > 5% | S046: 1,000<br>S047: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S048 / S049 | Average of eight learned forecasts | Estimated return > 5% | S048: 1,000<br>S049: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S054 | Average of eight learned forecasts | Estimated return > 5% | S054: 1% of current cash | Best odds<br>Cash available |
| S050 / S051 | Half combined forecast, half bookmaker | Estimated return > 5% | S050: 1,000<br>S051: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S059 | Team + chance quality + player form | Estimated return > 5% | S059: 1/4 Kelly; ≤2% cash | Best odds<br>Date cap 10% cash |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (5/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S060 | Bookmaker odds + team/chance/player form | Estimated return > 5% | S060: 1/4 Kelly; ≤2% cash | Best odds<br>Date cap 10% cash |
| S061 | Average of eight learned forecasts | Estimated return > 5% | S061: 1/4 Kelly; ≤2% cash | Best odds<br>Date cap 10% cash |
| S062 / S063 | Average of eight learned forecasts | Estimated return > 5%<br>Positive return: 6/8 models | S062: 1,000<br>S063: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| S064 | Average of eight learned forecasts | Estimated return > 5%<br>Only H | S064: 1,000 | Best odds<br>Cash available |
| S065 | Average of eight learned forecasts | Estimated return > 5%<br>Only D | S065: 1,000 | Best odds<br>Cash available |
| S066 | Average of eight learned forecasts | Estimated return > 5%<br>Only A | S066: 1,000 | Best odds<br>Cash available |
| S067 | Team + chance quality + player form | Estimated return > 10% | S067: 1,000 | Best odds<br>Cash available |
| S068 | Bookmaker odds + team/chance/player form | Estimated return > 10% | S068: 1,000 | Best odds<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (6/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S069 | Average of eight learned forecasts | Estimated return > 10% | S069: 1,000 | Best odds<br>Cash available |
| S070 | Team + chance quality + player form | Estimated return > 5%<br>Odds 1.5–5.0 | S070: 1,000 | Best odds<br>Cash available |
| S071 | Bookmaker odds + team/chance/player form | Estimated return > 5%<br>Odds 1.5–5.0 | S071: 1,000 | Best odds<br>Cash available |
| S072 | Average of eight learned forecasts | Estimated return > 5%<br>Odds 1.5–5.0 | S072: 1,000 | Best odds<br>Cash available |
| S073 | Original SVM forecast | Estimated return > 0% | S073: 1,000 | Best odds<br>Cash available |
| S074 | Original SVM forecast | Estimated return > 2.5% | S074: 1,000 | Best odds<br>Cash available |
| S075 | Original SVM forecast | Estimated return > 7.5% | S075: 1,000 | Best odds<br>Cash available |
| S076 | Original SVM forecast | Estimated return > 10% | S076: 1,000 | Best odds<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (7/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| S077 | Original SVM forecast | Estimated return > 15% | S077: 1,000 | Best odds<br>Cash available |
| S078 | Original SVM forecast | Estimated return > 20% | S078: 1,000 | Best odds<br>Cash available |
| S079 | Original neural-network forecast | Estimated return > 0% | S079: 1,000 | Best odds<br>Cash available |
| S080 | Original neural-network forecast | Estimated return > 2.5% | S080: 1,000 | Best odds<br>Cash available |
| S081 | Original neural-network forecast | Estimated return > 7.5% | S081: 1,000 | Best odds<br>Cash available |
| S082 | Original neural-network forecast | Estimated return > 10% | S082: 1,000 | Best odds<br>Cash available |
| S083 | Original neural-network forecast | Estimated return > 15% | S083: 1,000 | Best odds<br>Cash available |
| S084 | Original neural-network forecast | Estimated return > 20% | S084: 1,000 | Best odds<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (8/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| P001 / P002 | Team form (logistic) | Estimated return > 5% | P001: 1,000<br>P002: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P003 / P004 | Team form (logistic) | Estimated return > 5% | P003: 1,000<br>P004: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| P005 / P006 | Team form (boosted trees) | Estimated return > 5% | P005: 1,000<br>P006: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P007 / P008 | Team form (boosted trees) | Estimated return > 5% | P007: 1,000<br>P008: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| P009 / P010 | Player form (logistic) | Estimated return > 5% | P009: 1,000<br>P010: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P011 / P012 | Player form (logistic) | Estimated return > 5% | P011: 1,000<br>P012: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| P013 / P014 | Player form (boosted trees) | Estimated return > 5% | P013: 1,000<br>P014: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P015 / P016 | Player form (boosted trees) | Estimated return > 5% | P015: 1,000<br>P016: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes

# EPL appendix: strategy rules (9/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

| Strategy IDs | Data / model | When it bets | Requested money per bet | Odds / total limit |
|---|---|---|---|---|
| P017 / P018 | Player form + estimated XI/depth (logistic) | Estimated return > 5% | P017: 1,000<br>P018: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P019 / P020 | Player form + estimated XI/depth (logistic) | Estimated return > 5% | P019: 1,000<br>P020: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| P021 / P022 | Player form + estimated XI/depth (trees) | Estimated return > 5% | P021: 1,000<br>P022: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P023 / P024 | Player form + estimated XI/depth (trees) | Estimated return > 5% | P023: 1,000<br>P024: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| P025 / P026 | Team + player form/XI/depth (logistic) | Estimated return > 5% | P025: 1,000<br>P026: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P027 / P028 | Team + player form/XI/depth (logistic) | Estimated return > 5% | P027: 1,000<br>P028: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |
| P029 / P030 | Team + player form/XI/depth (trees) | Estimated return > 5% | P029: 1,000<br>P030: 1/4 Kelly; ≤2% cash | Best odds<br>Cash available |
| P031 / P032 | Team + player form/XI/depth (trees) | Estimated return > 5% | P031: 1,000<br>P032: 1/4 Kelly; ≤2% cash | Bet365<br>Cash available |

Sources: all_strategy_catalog.json | Full paths in notes